EE3032 Introduction to VLSI Design—Homework 2

1. Fig. 1(a) shows an inverter driving a capacitive load C_L. If two input stimuli as shown in Fig. 1(b) are applied, the fall time of the output Vout for the input stimulus a is shorter than that for the input stimulus b. Why? Explain the reasons. (3 points)

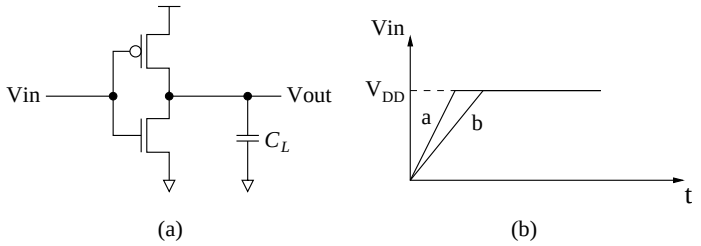


Figure 1: (a) An inverter drives a load C_L. (b) Two input waveforms with different rise times.

2. Determine the worst case high-to-low and low-to-high propagation delays at nodes F and Z for the circuit shown in Fig. 2 using Elmore delay model. Assume that the PMOS and NMOS transistors have effective channel resistance R_p and R_n, respectively. Also, C_F and C_Z are the total capacitance at nodes F and Z, respectively. (2 points)

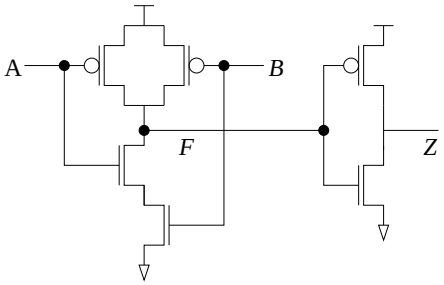
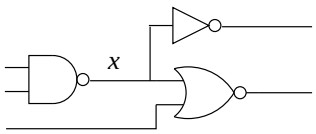


Figure 2: Two-input NAND gate driving an inverter.

3. Assume that the size of the two transistors of an inverter is as follows: M1=$W_p/L_p = 4$ and M2=$W_n/L_n = 2$. If Vin=$\frac{1}{2}V_{DD}$, the Vout=$\frac{1}{2}V_{DD}$. Calculate the ratio of the μ_n to μ_p. (2 points)

4. Fig. 4 shows a simple logic circuit. Assume that the gate capacitance of the invert is C_g; the β value of the PMOS and NMOS of the inverter is β_p and β_n, respectively. If the equivalent β of PMOS network and NMOS network of the two-input NOR gate is β_{peq} and β_{neq}, respectively. Assume that $\beta_{peq} = \beta_p$ and $\beta_{neq} = \beta_n$; $\mu_n = 2\mu_p$. Calculate the capactiance of the node x without considering the parasitic capacitances of wire and drain. (3 points)